

Education

- Birla Institute of Technology, Mesra

2023 – 2027
- Bachelor of Technology in Computer Science and Engineering (CGPA: 8.42 / 10.0)

Ranchi, India

Relevant Coursework: Data Structures & Algorithms (DSA), Operating Systems, Database Management Systems (DBMS), Systems Design, Artificial Intelligence and Machine Learning (AIML), Software Engineering, Soft Computing.

Achievements

- **Core AI Repository Maintainer**,Biscuit – Author and maintainer of an open-source IDE ecosystem with 250+ stars
- **Top 10 Finish Monad Blitz**,Built an AI-powered escrow platform using Sarvam AI and Monad Blockchain; placed 8th out of 50+ teams.
- **OSS Contributor**,Zulip – Shipped multiple layout features to the production codebase of an enterprise open-source system.
- **LeetCode Knight Badge**,Secured global ranking top 5% (Max rating: 1,893); engineered solutions for 400+ complex problems.
- **Data Automation Challenge Runner-up**,Engineered a semantic search and reasoning engine using localized vector embeddings.

Work Experience

- AI Engineer Intern (Full Stack)

July 2025 – Sept 2025
- Hooman Digital LLP

Remote, India
- Architected an **AI-native multimodal reasoning pipeline** leveraging LLMs to automate chart trend analysis, successfully reducing manual operational workflow overhead by **65%**.
 - Built a provider-agnostic LLM orchestration framework in Python with deterministic fallback logic, ensuring zero loss of contextual integrity during token rate-limit outages.
 - Optimized asynchronous real-time streaming pipelines via Server-Sent Events, boosting event propagation throughput by **40%** while keeping execution log rendering latency under **45ms**.
- Full Stack Engineer Intern

Nov 2025 – Feb 2026
- Morvion

Remote, Zurich
- Initialized a modular SaaS backend platform using NestJS, implementing an automated cache-eviction strategy that lowered high concurrency database connection locks by **35%** across enterprise loops.
 - Engineered high-throughput asynchronous data ingestion pipelines via third-party API integrations, saving over **10+ hours weekly** of manual data entry while processing thousands of dense monthly accounts.
 - Deployed background workflow orchestration suites with automated token-refresh routines, achieving an estimated **25% reduction in boilerplate overhead** across cloud transaction networks.
- Deep Learning Research Intern

May 2025 – July 2025
- NIT Calicut (Advisor: Dr. S.K. Behera)

Calicut, India
- Developed a hybrid Spatial CNN + Temporal Transformer architecture for biometric signal classification, improving performance by **15%** over baseline models.
 - Designed custom spatial-temporal positional encoding mechanisms to preserve structural relationships within transformer attention layers.
 - Built optimized PyTorch preprocessing and training pipelines, reducing end-to-end experimentation time by over **25%**.

Projects

- Biscuit – Open Source Agentic IDE (Integrated Development Environment)

Python, LangChain, Claude, Ollama, Agentic AI Systems, Tool Calling, Context Engineering

Source Code
- Built an open-source agentic IDE enabling autonomous repository understanding, code generation, debugging, planning, and multi-step software engineering workflows through tool-calling AI agents.
 - Designed agent workflows using ReAct, structured outputs, MCP integrations, and **human-in-the-loop feedback loops** to improve reliability of long-running coding tasks.
 - Implemented repository-scale code intelligence using Tree-sitter, Language Server Protocol (LSP), and Debug Adapter Protocol (DAP), enabling semantic understanding across **20+** programming languages.
 - Designed an extensible plugin architecture enabling third-party developers to build and distribute custom AI tools, workflows, and integrations.
 - Built global indexing, retrieval, and context-engineering systems delivering sub-**100ms** search across repositories exceeding **1M+** lines of code.
- Ted – Multi-Agent Orchestration Platform Infrastructure

Rust, Tauri, React, Google Cloud Platform (GCP), GitHub Actions, IPC Architecture

Source Code
- Built a cross-platform orchestration platform in Rust for coordinating distributed AI agents and developer workflows.
 - Engineered an agentic browser integration that allowed AI agents to test applications, inspect UI state, verify code changes, and interact with localhost environments as part of autonomous development loops.
 - Developed real-time observability tooling for monitoring agent execution, coordination, and telemetry across distributed systems.

Technical Skills

- **AI:** Agentic AI, Tool Calling, Multi-Agent Systems, LangGraph, LangChain, Model Context Protocol (MCP), RAG, Context Engineering, Prompt Engineering, LLM Evaluation, Semantic Search, Anthropic API, OpenAI API, PyTorch
- **Programming Languages:** Python, Rust, TypeScript, C/C++, SQL
- **Backend & Infra:** FastAPI, Django, REST APIs, WebSockets, Microservices, gRPC, PostgreSQL, Redis, Docker, CI/CD, Linux/Shell
- **Frontend:** React, Next.js, Tauri, HTML/CSS, Tailwind CSS
- **Developer Tooling:** Agent Client Protocol (ACP), Language Server Protocol (LSP), Debug Adapter Protocol (DAP), Tree-sitter
- **Databases:** PostgreSQL, pgvector, Redis, Pinecone, Vector Search Systems